Lecture: Alternates for flexibility

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Agenda

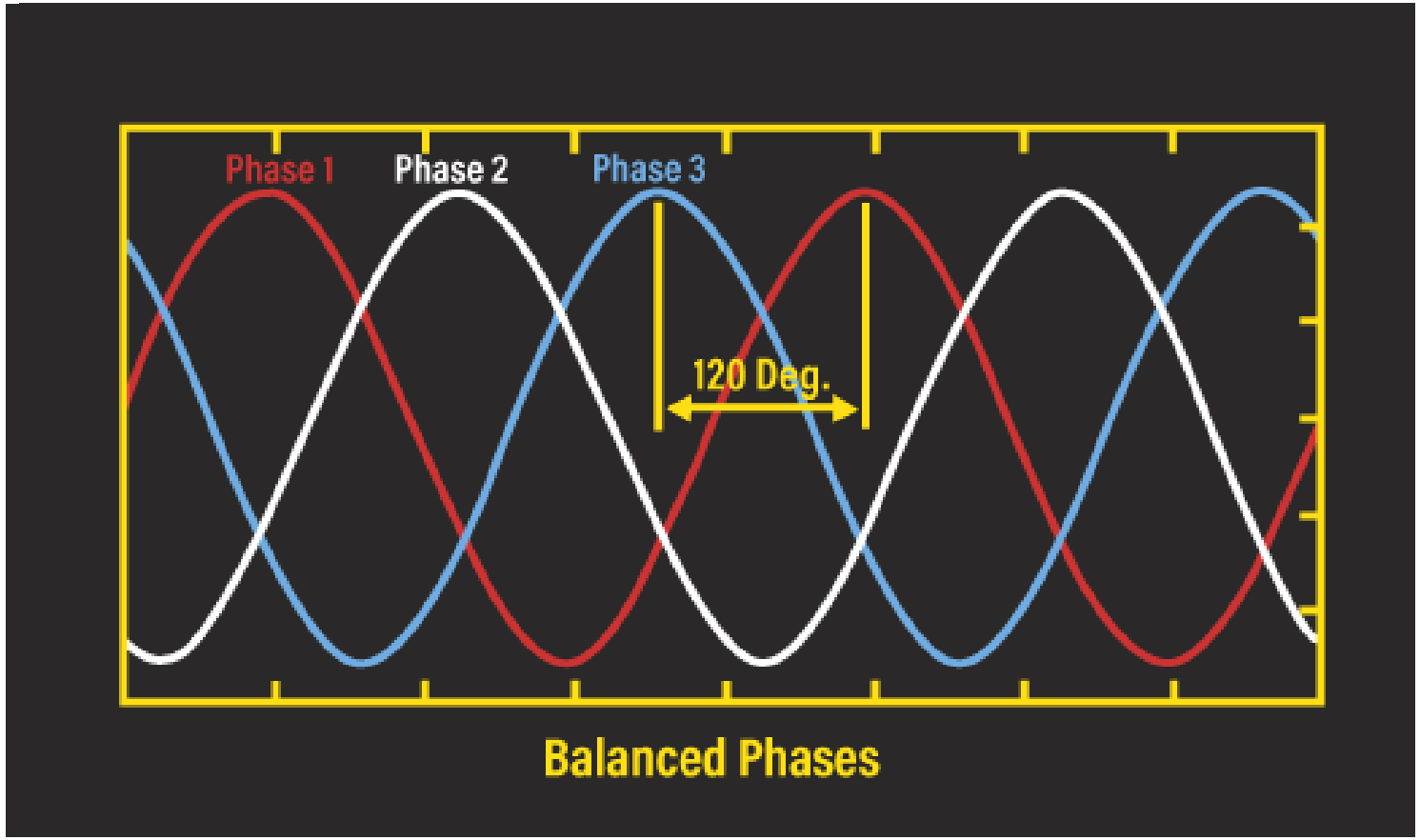
- Solving distribution network issues
- Alternate solutions
  - Non-intrusive solutions
  - Policy/Market driven
  - Active devices
  - Network upgrades
  - Curtailment/load shedding
- Conclusions

Solving distribution network incidents

Distribution network incidents

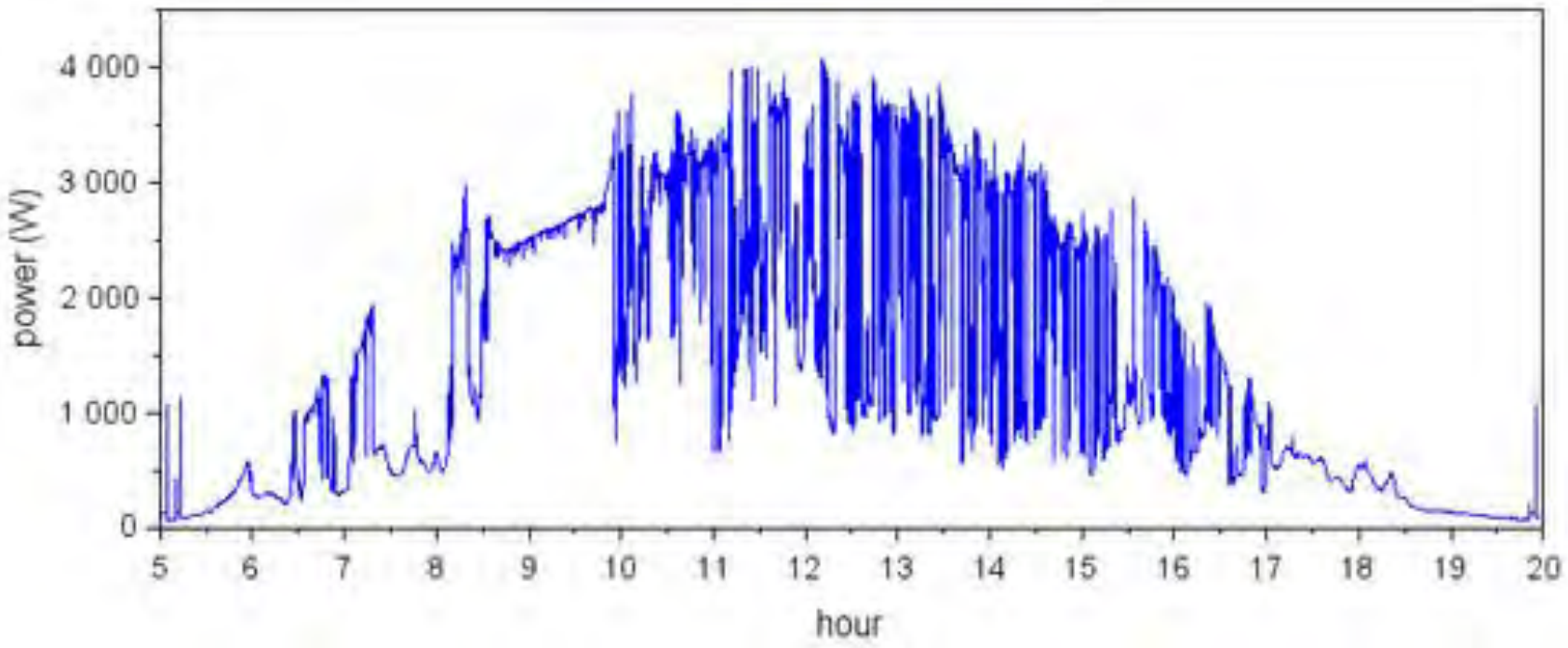
- Steady state network incidents
  - Evolve in minutes to hours
  - Includes thermal overloads, voltage violations, unbalance etc.
- Transient / Dynamic network incidents
  - Evolve in milliseconds to seconds
  - Includes voltage sags/swells, flicker, frequency dips, transient over-voltages, oscillations, etc.
- Short-circuit, Fault localization and mitigation
  - high-impact incidents involving abrupt faults that cause very high fault currents and immediate voltage collapse at the fault location
- Weather induced incidents – cyclone, hurricane, storms, …
  - Includes cyclone/hurricane, lightning, flooding, ice/snow loading etc.

Voltage unbalance – a steady state problem



Source: https://www.energymanagermagazine.co.uk/the-importance-of-not-losing-your-balance/

Voltage Fluctuations and Flicker in Prosumer PV Installation



Source: https://www.mdpi.com/1996-1073/15/6/2075

Line to ground fault post storm in Texas

Source: https://blog.ucs.org/guest-commentary/the-electricity-distribution-system-can-stay-resilient-in-extreme-weather/

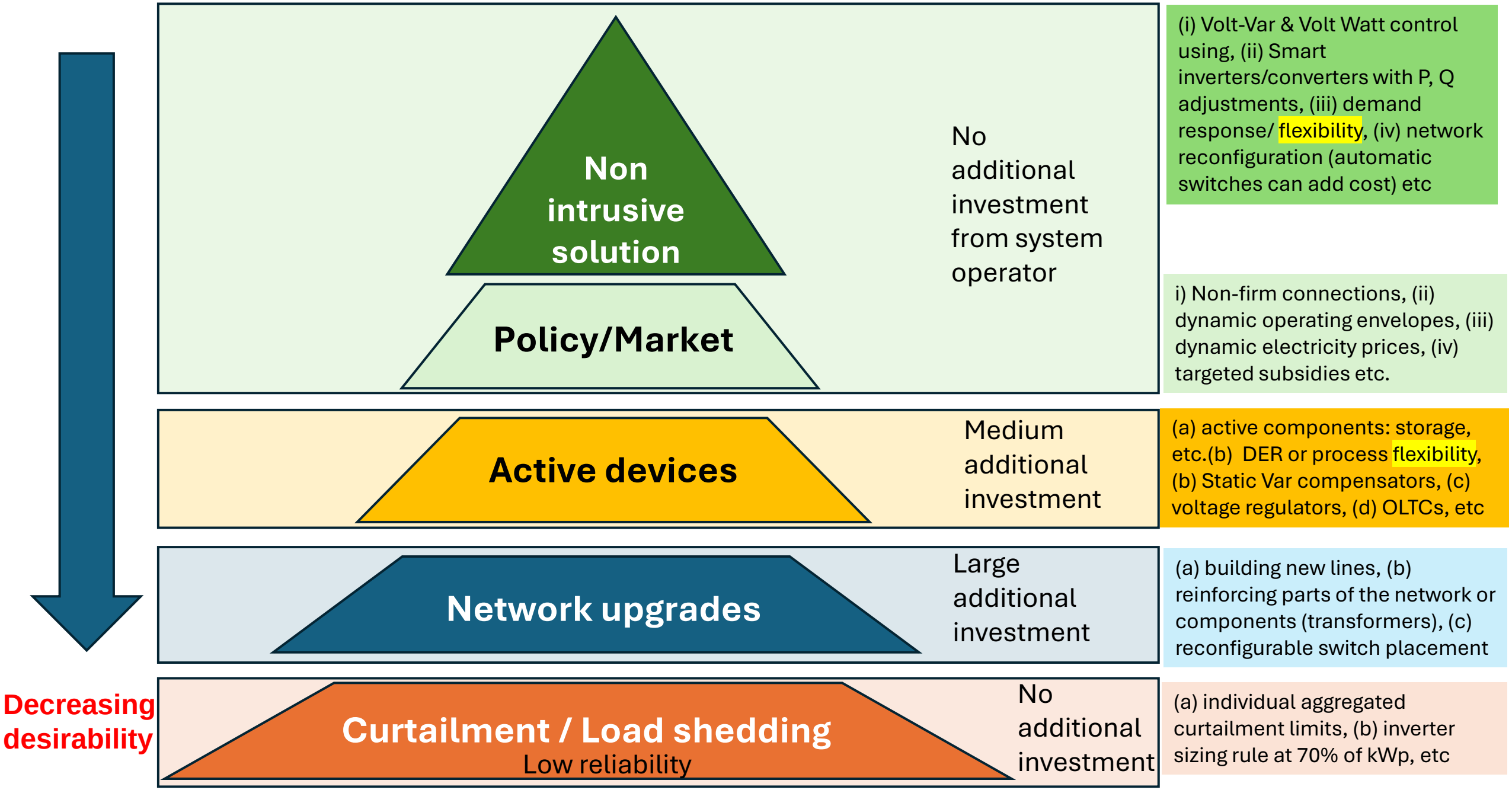
Short-circuit, Fault localization and mitigation

These are high-impact incidents involving abrupt faults that cause very high fault currents and immediate voltage collapse at the fault location. They require rapid protection response. Examples include:

- Phase-to-phase faults, Single-line-to-ground faults (most common), Three-phase bolted faults (rare but highly severe).
- Broken conductor incidents, creating hazardous touch and step voltages.
- Arc faults inside switchgear, transformers, or cable joints.

Mitigation: Protection system activation (fuses, reclosers, relays, sectionalizers).

Alternate solutions in distribution network


Non intrusive solution
Policy/Market
No additional investment from system operator
(i) Volt-Var & Volt Watt control using, (ii) Smart inverters/converters with P, Q adjustments, (iii) demand response/ flexibility, (iv) network reconfiguration (automatic switches can add cost) etc
i) Non-firm connections, (ii) dynamic operating envelopes, (iii) dynamic electricity prices, (iv) targeted subsidies etc.
Active devices
Medium additional investment
(a) active components: storage, etc.(b) DER or process flexibility, (b) Static Var compensators, (c) voltage regulators, (d) OLTCs, etc
Network upgrades
Large additional investment
(a) building new lines, (b) reinforcing parts of the network or components (transformers), (c) reconfigurable switch placement
Curtailment / Load shedding
Low reliability
No additional investment
(a) individual aggregated curtailment limits, (b) inverter sizing rule at 70% of kWp, etc
Decreasing desirability

Solution – 1: Non-intrusive solutions

Smart inverters

- Reactive power capability of smart inverters ---voltage regulation in LV distribution networks

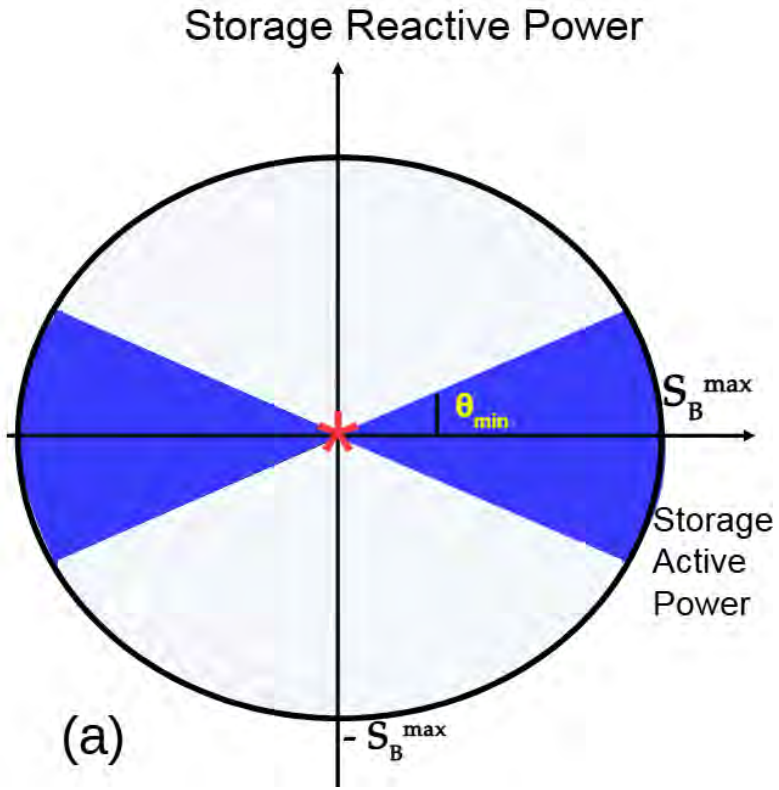


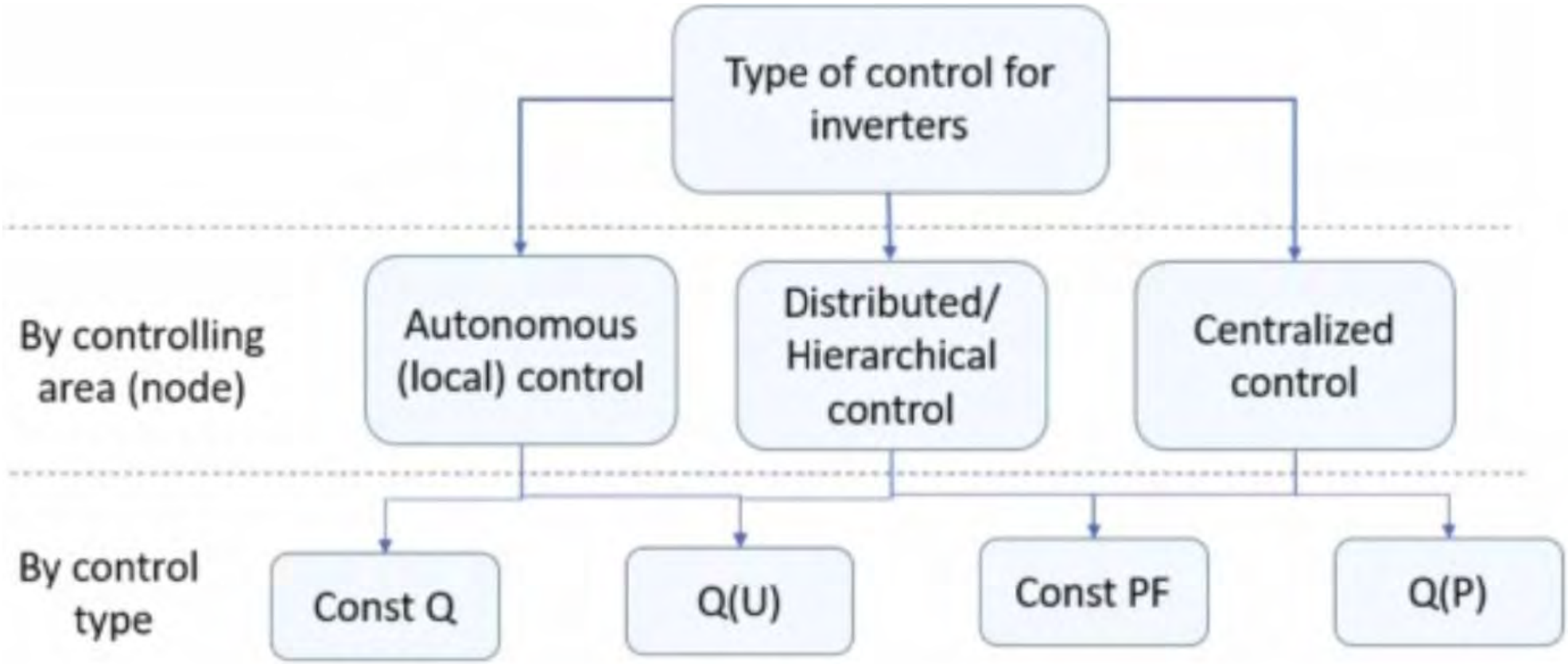
Fig. Inverter capability curve for energy storage

Desired capabilities of smart inverters according to IEEE-1547-2018

Performance Categories	Mandatory voltage regulation capability modes				
	Constant power factor ($\cos\phi$)	Constant reactive power	Voltage-reactive power (volt-Var)	Active power-reactive power	Voltage-active power (volt-watt)
Category A: Minimum performance	yes	yes	yes	Not required	Not required
Category B: for smart inverters	yes	yes	yes	yes	yes

Source: Can locational disparity of prosumer energy optimization due to inverter rules be limited?

Types of control of smart inverters



Demand side management

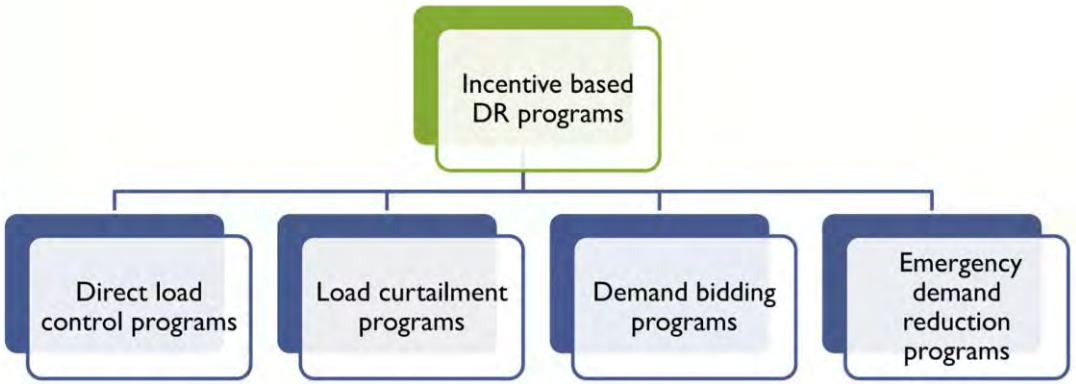
- Demand response(DR) is defined as a change in the electricity consumption of a site to achieve a new consumption profile, e.g. decreasing a consumption profile at a time where there is not enough energy generation in the grid.

- Main goal is to encourage consumers to use less energy during peak hours, or to move the time of energy use to off-peak hours

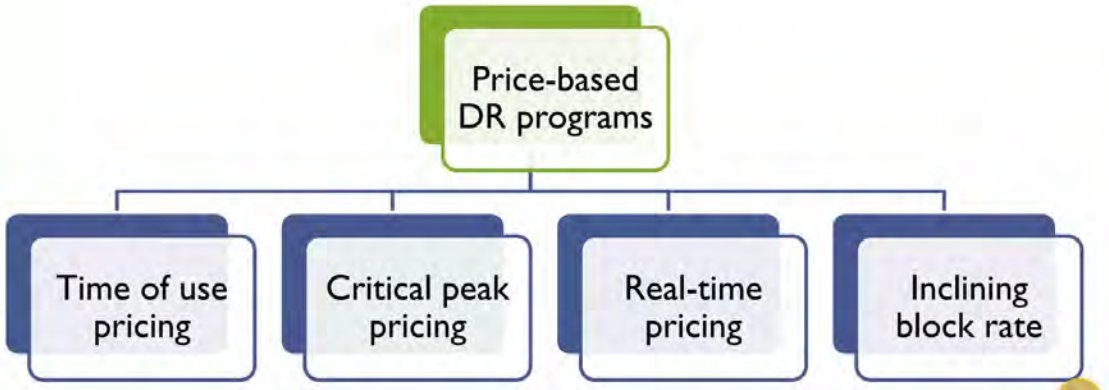
- An example is to use of energy storage units to store energy during off-peak hours and discharge them during peak hours.

- Electric boilers, EV charging, Smart home appliances

Demand Response (DR) programs



These programs pay participating consumers, **who reduce their consumption at peak hours or during events.**

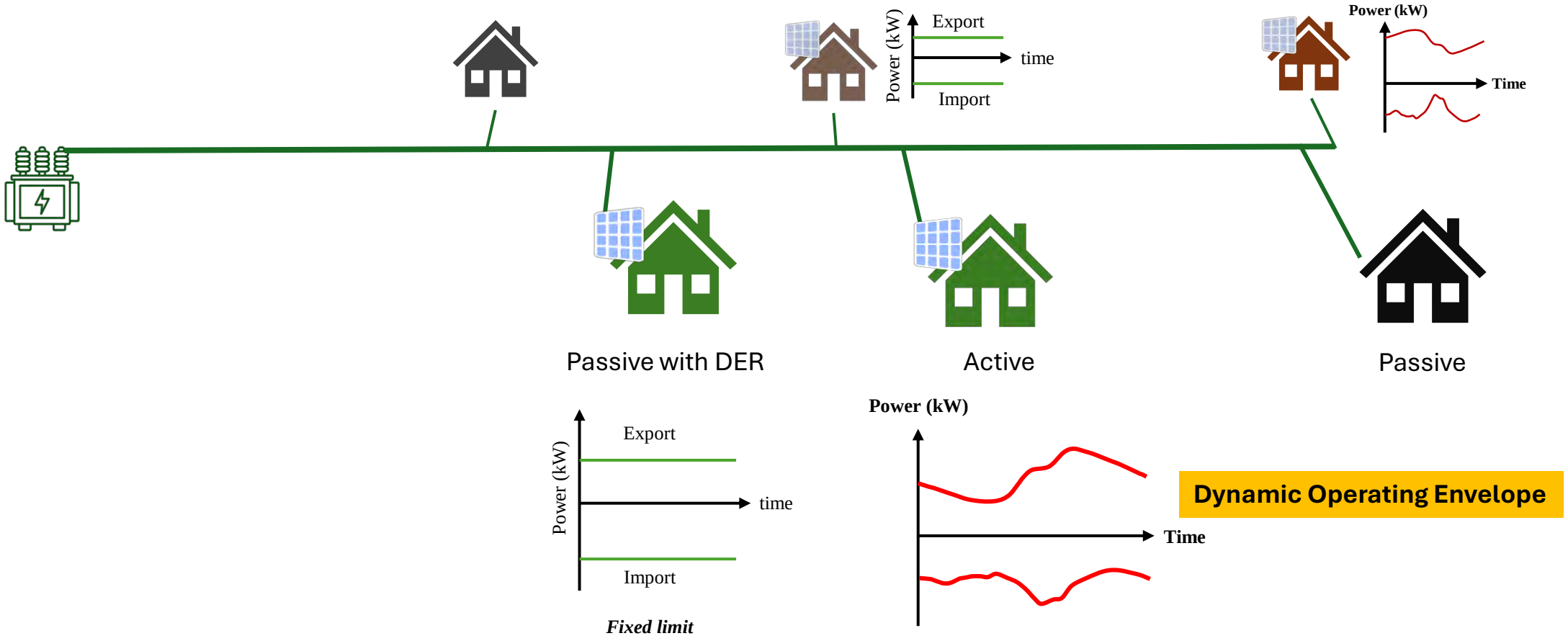


In price-based DR programs, **the consumers are charged with different prices at different times of consumption.** In this way, the consumers are charged according to the supply cost of electricity. By increasing tariffs during peak demand hours and contingencies, utilities incentivise consumers to reduce their consumption.

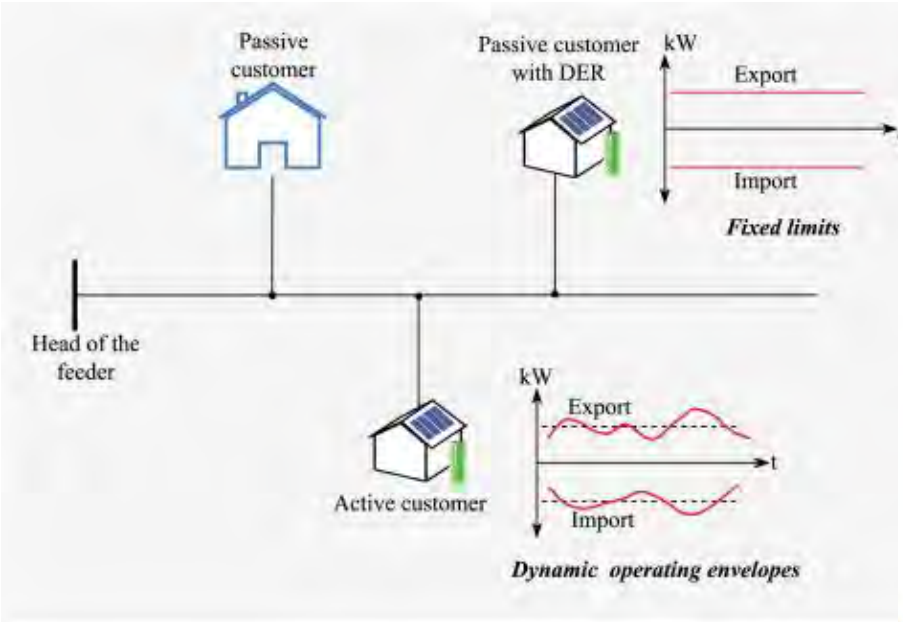
Solution – 2: Policy/Market driven

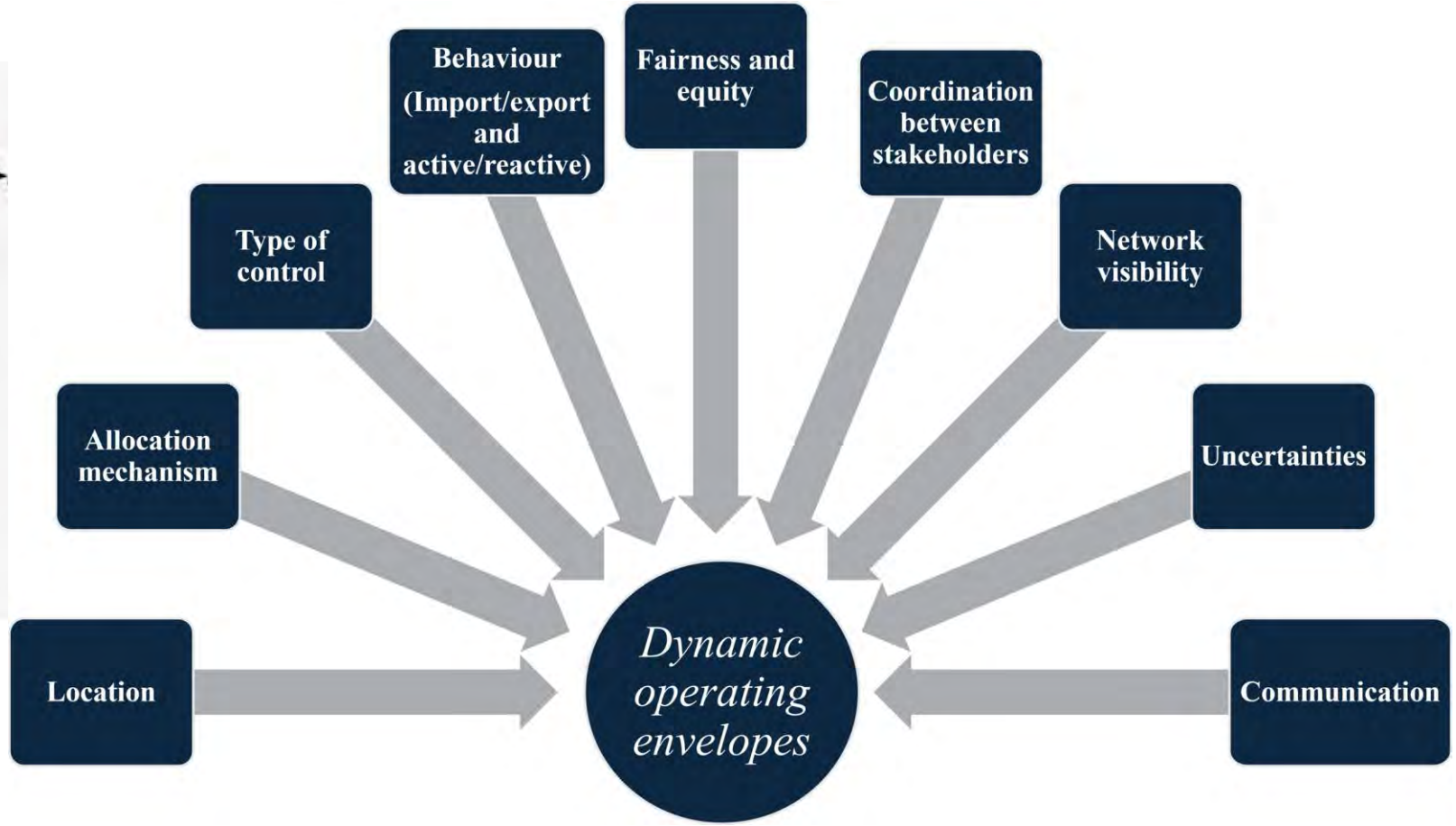
Non-firm connections

- Flexible grid access instead of guaranteed firm capacity
- During network congestion or voltage violations, the network operator may **temporarily limit or curtail power injection or consumption**



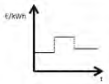
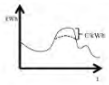
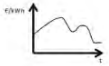
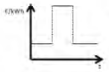
Dynamic operating envelops





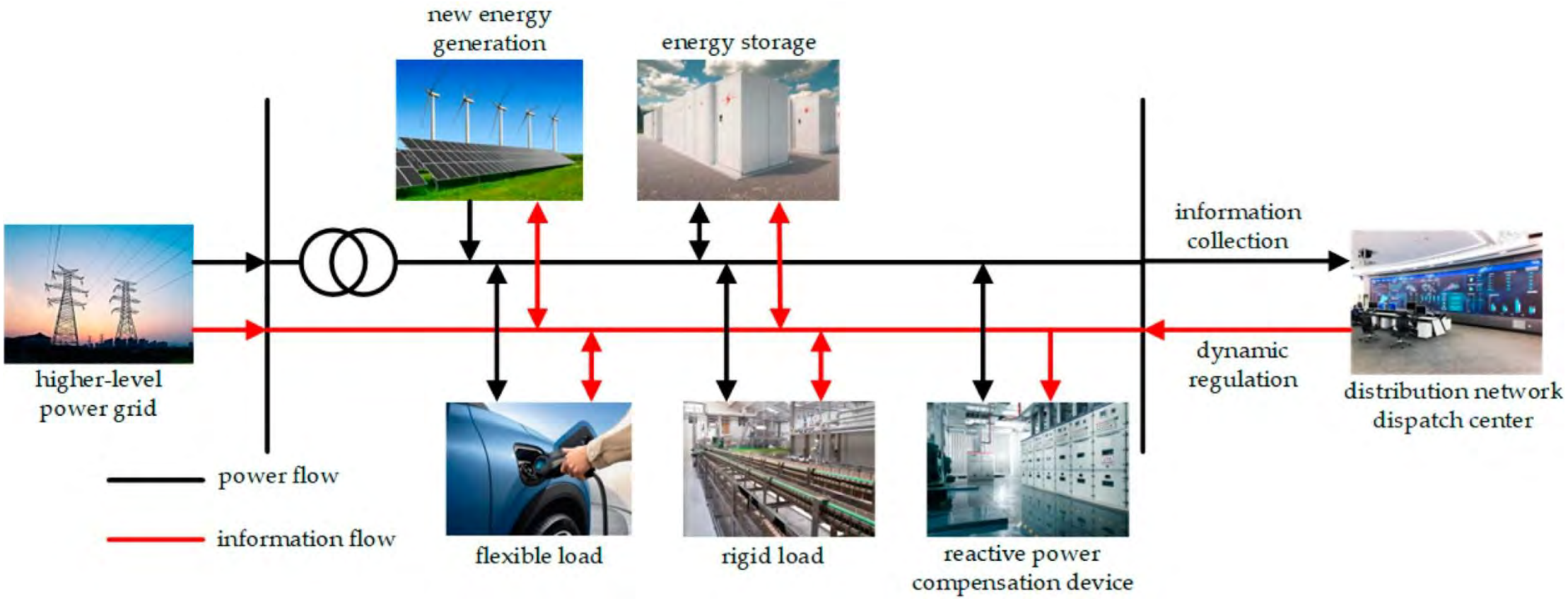
Dynamic electricity prices

Possible dynamic pricing options for DER management

Basic Dynamic Pricing Options			Specific incentives for baseline adjustment		
Time-Of-Use (TOU)	Fixed electricity prices for different time blocks within a time period		**Peak time rebates (PTR)**	A rebate when electricity is reduced compared to baseline consumption, within certain hours in a year.	
Real-Time-Pricing (RTP)	An hourly rate depending on the day ahead real-time price of electricity		**Interruptible Capacity Program (ICAP), Interruptible load**	A rebate when electricity is reduced below a baseline value.	
Critical Peak Pricing (CPP)	High electricity price periods for certain (fixed) days of time within a year		**Emergency Demand Response**	Mandatory commitment to reduce load, with penalties if not supplied.	

Solution – 3: Active devices

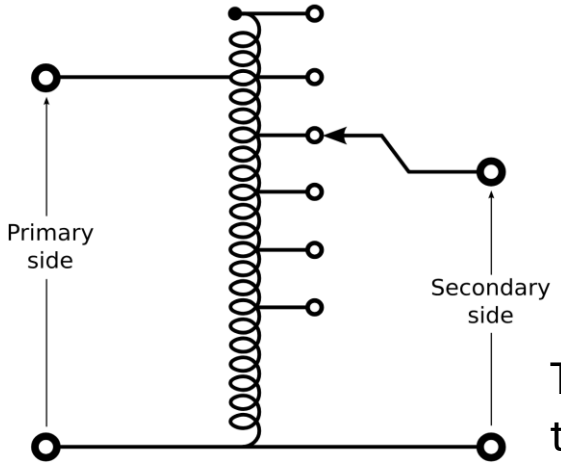
Active networks with DERs



Active devices

Energy storage

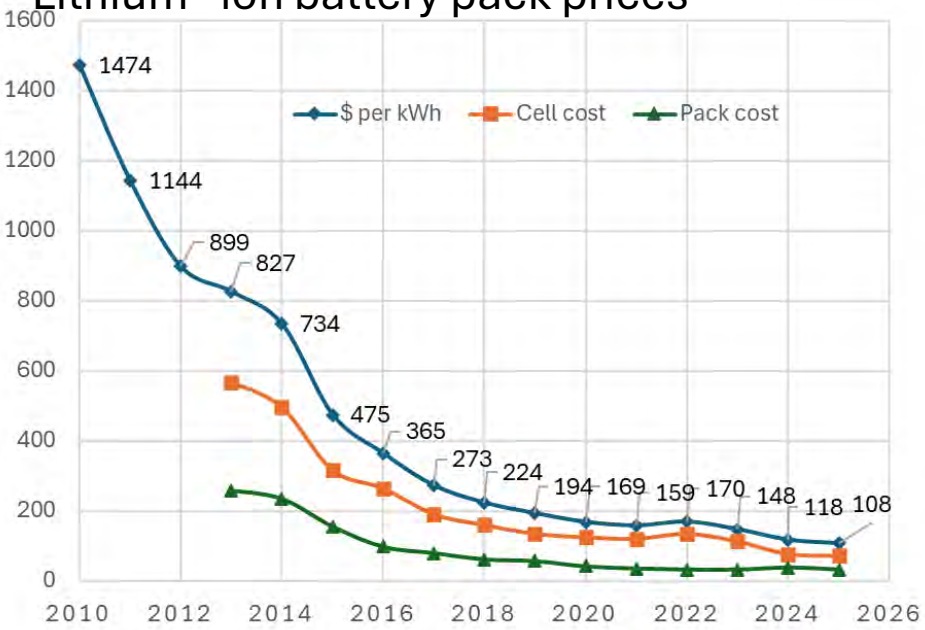
Voltage regulator



Tap changing transformer

community energy storage systems

Lithium –ion battery pack prices



Source: BloombergNEF 2025

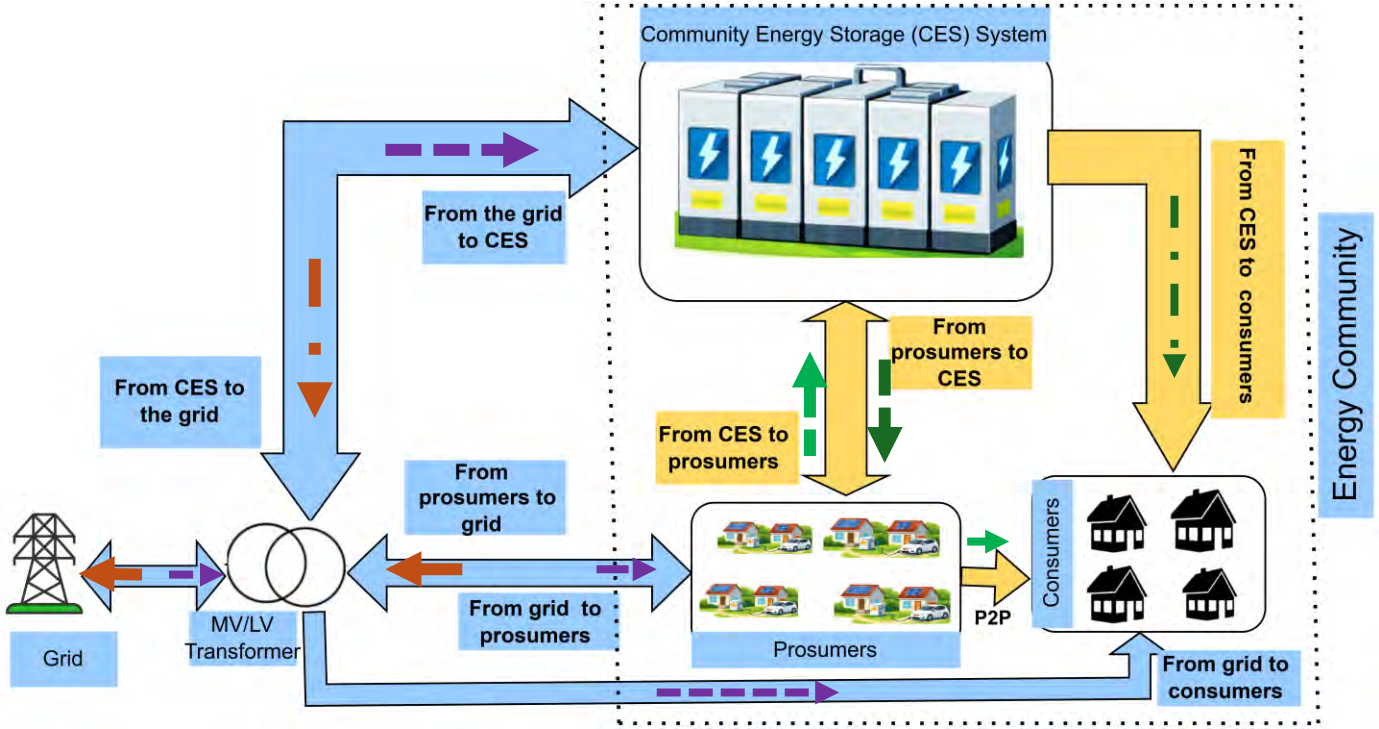


Fig.2. CES system configuration within local energy community

Solution – 4: Network upgrades

Network upgrades

- Building new distribution lines
- Change feeder topology to redistribute power flows and voltage profiles.
- Useful where alternative paths already exist and switch states can be changed safely.
- Automatic switching adds cost; manual/scheduled reconfiguration is cheaper but slower.
- Requires protection coordination and operational validation.

Solution – 5: Curtailment/Load shedding

What is curtailment?

- Curtailment is when the system operator restricts the ability of generators to export their power or reduce their load
- Is a last-resort measure to maintain stability

References

- Gelazanskas L. Demand side management in smart grid: A review and proposals for future direction. Sustainable cities and socities
- Gayan Lankeshwara , "A review on network-aware control of distributed energy resources under the dynamic operating envelopes framework", Renewable and Sustainable Energy Reviews
- Cherrelle Eid, "Managing electric flexibility from Distributed Energy Resources: A review of incentives for market design", Renewable and Sustainable Energy Reviews

Course acknowledgements

Kennis kent geen einde

IMPROCAP

https://www.improcap.eu/

DiGiRES

https://www.digi-res.eu/

C2-FlexIQ

https://research.kuleuven.be/portal/nl/project/3E240512

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Thank you for your attention!

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Demand response and network reconfiguration

Demand response

- Shift flexible demand to times/locations with excess generation ,lower stress, or lower price.
- Examples: EV charging, water heaters, HVAC, storage.

Need for alternative flexibility solutions

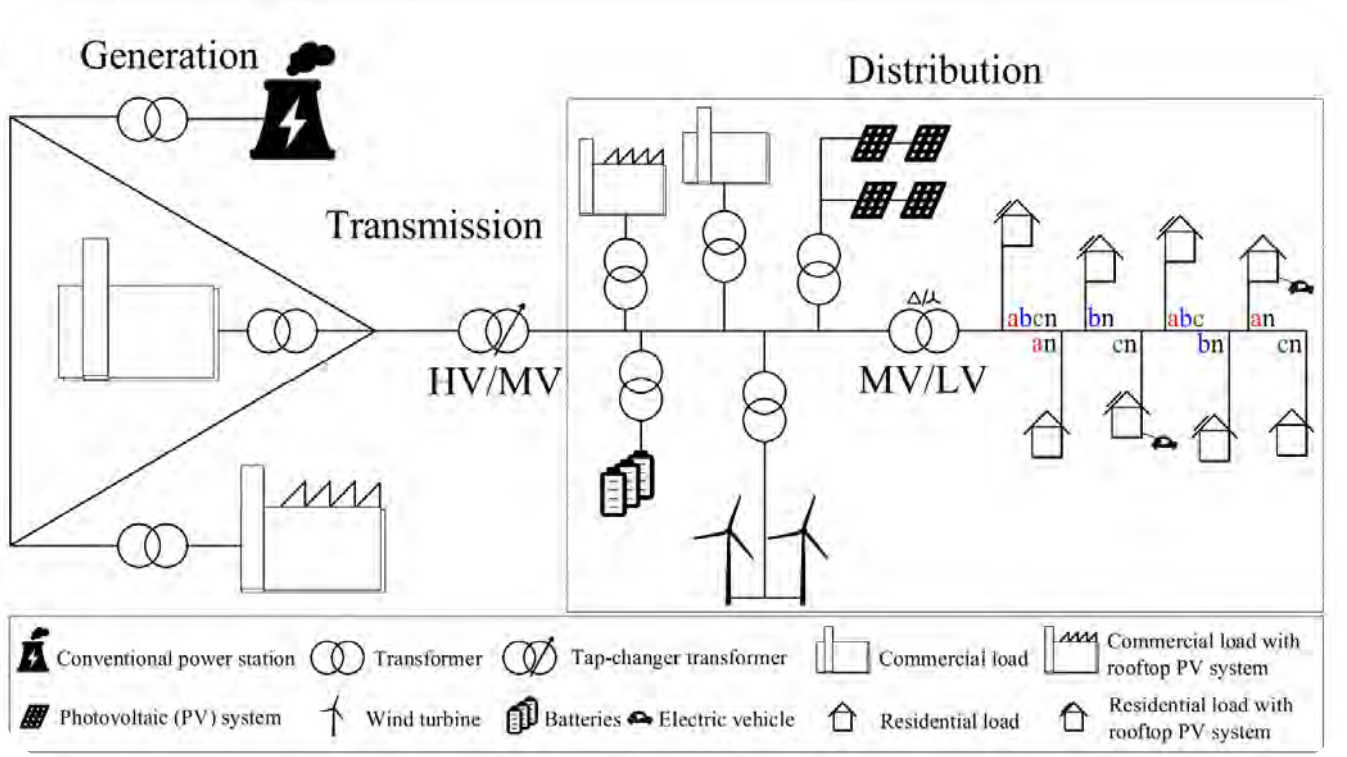

Fig.1. A one-line diagram of distribution network as part of power systems

- Shift from passive → active distribution (Bidirectional Flows)
- High distributed energy resources (DER) integration in power networks
- PV, BESS, EVs, HP, HVAC, and Smart Loads

Challenges in power system due to DERs

- Over/undervoltage, congestion, coordination, and protection, grid upgrade costs